

SMSfileBased

- |
- |— Student.java
- |— StudentDatabase.java
- |— Main.java
- |— Student.txt

//Main.java

```
import java.util.Scanner;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        StudentDatabase db = new StudentDatabase();

        while (true) {

            System.out.println("\n===== STUDENT MANAGEMENT SYSTEM =====");
            System.out.println("1. Add Student");
            System.out.println("2. View Students");
            System.out.println("3. Search Student");
            System.out.println("4. Update Student");
            System.out.println("5. Delete Student");
            System.out.println("6. Exit");
            System.out.print("Enter Your Choice: ");

            int choice = sc.nextInt();

            switch (choice) {

                case 1:

                    System.out.print("Enter ID: ");
                    int id = sc.nextInt();
                    sc.nextLine();

                    System.out.print("Enter Name: ");
                    String name = sc.nextLine();

                    System.out.print("Enter Age: ");
                    int age = sc.nextInt();
                    sc.nextLine();

                    System.out.print("Enter Course: ");
                    String course = sc.nextLine();

                    Student student = new Student(id, name, age, course);

                    db.addStudent(student);

                    break;
```

case 2:

```
db.viewStudents();
```

```
break;
```

case 3:

```
System.out.print("Enter Student ID: ");  
id = sc.nextInt();
```

```
db.searchStudent(id);
```

```
break;
```

case 4:

```
System.out.print("Enter Student ID: ");  
id = sc.nextInt();  
sc.nextLine();
```

```
System.out.print("Enter New Name: ");  
name = sc.nextLine();
```

```
System.out.print("Enter New Age: ");  
age = sc.nextInt();  
sc.nextLine();
```

```
System.out.print("Enter New Course: ");  
course = sc.nextLine();
```

```
Student updateStudent = new Student(id, name, age, course);
```

```
db.updateStudent(updateStudent);
```

```
break;
```

case 5:

```
System.out.print("Enter Student ID: ");  
id = sc.nextInt();
```

```
db.deleteStudent(id);
```

```
break;
```

case 6:

```
System.out.println("Thank You!");  
sc.close();  
System.exit(0);
```

default:

```
System.out.println("Invalid Choice!");
```

```
}  
}}}
```

//Student.java

```
public class Student {

    private int id;
    private String name;
    private int age;
    private String course;

    public Student(int id, String name, int age, String course) {
        this.id = id;
        this.name = name;
        this.age = age;
        this.course = course;
    }

    public int getId() {
        return id;
    }

    public String getName() {
        return name;
    }

    public int getAge() {
        return age;
    }

    public String getCourse() {
        return course;
    }

    public void setId(int id) {
        this.id = id;
    }

    public void setName(String name) {
        this.name = name;
    }

    public void setAge(int age) {
        this.age = age;
    }

    public void setCourse(String course) {
        this.course = course;
    }

    @Override
    public String toString() {
        return id + "," + name + "," + age + "," + course;
    }
}
```

//StudentDatabase.java

```
import java.io.*;
import java.util.ArrayList;

public class StudentDatabase {

    private final String FILE_NAME = "Student.txt";

    // Add Student
    public void addStudent(Student student) {

        try (BufferedWriter bw = new BufferedWriter(new FileWriter(FILE_NAME, true))) {

            bw.write(student.getId() + "," +
                    student.getName() + "," +
                    student.getAge() + "," +
                    student.getCourse());

            bw.newLine();

            System.out.println("Student Added Successfully!");

        } catch (IOException e) {
            System.out.println("Error : " + e.getMessage());
        }
    }

    // View Students
    public void viewStudents() {

        try (BufferedReader br = new BufferedReader(new FileReader(FILE_NAME))) {

            String line;

            while ((line = br.readLine()) != null) {

                String[] data = line.split(",");

                System.out.println("-----");
                System.out.println("ID : " + data[0]);
                System.out.println("Name : " + data[1]);
                System.out.println("Age : " + data[2]);
                System.out.println("Course : " + data[3]);
            }

        } catch (IOException e) {
            System.out.println("Error : " + e.getMessage());
        }
    }

    // Search Student
    public void searchStudent(int id) {

        try (BufferedReader br = new BufferedReader(new FileReader(FILE_NAME))) {

            String line;
            boolean found = false;
```

```

while ((line = br.readLine()) != null) {

    String[] data = line.split(",");

    if (Integer.parseInt(data[0]) == id) {

        System.out.println("-----");
        System.out.println("ID   : " + data[0]);
        System.out.println("Name  : " + data[1]);
        System.out.println("Age   : " + data[2]);
        System.out.println("Course : " + data[3]);

        found = true;
        break;
    }
}

if (!found) {
    System.out.println("Student Not Found!");
}

} catch (IOException e) {
    System.out.println("Error : " + e.getMessage());
}
}

// Update Student
public void updateStudent(Student student) {

    ArrayList<String> list = new ArrayList<>();

    try (BufferedReader br = new BufferedReader(new FileReader(FILE_NAME))) {

        String line;
        boolean found = false;

        while ((line = br.readLine()) != null) {

            String[] data = line.split(",");

            if (Integer.parseInt(data[0]) == student.getId()) {

                list.add(student.getId() + "," +
                    student.getName() + "," +
                    student.getAge() + "," +
                    student.getCourse());

                found = true;

            } else {

                list.add(line);
            }
        }

        BufferedWriter bw = new BufferedWriter(new FileWriter(FILE_NAME));
    }
}

```

```

        for (String s : list) {
            bw.write(s);
            bw.newLine();
        }

        bw.close();

        if (found)
            System.out.println("Student Updated Successfully!");
        else
            System.out.println("Student Not Found!");

    } catch (IOException e) {
        System.out.println("Error : " + e.getMessage());
    }
}

// Delete Student
public void deleteStudent(int id) {

    ArrayList<String> list = new ArrayList<>();

    try (BufferedReader br = new BufferedReader(new FileReader(FILE_NAME))) {

        String line;
        boolean found = false;

        while ((line = br.readLine()) != null) {

            String[] data = line.split(",");

            if (Integer.parseInt(data[0]) == id) {

                found = true;

            } else {

                list.add(line);
            }
        }

        BufferedWriter bw = new BufferedWriter(new FileWriter(FILE_NAME));

        for (String s : list) {
            bw.write(s);
            bw.newLine();
        }

        bw.close();

        if (found)
            System.out.println("Student Deleted Successfully!");
        else
            System.out.println("Student Not Found!");

    } catch (IOException e) {
        System.out.println("Error : " + e.getMessage());    } }

```

// Student.txt (Student data will be automatically saved here when you enter data through the console.)

11,Bunny,23,MCA
12,Pranali,22,MCA
13,Ketaki,21,Arch
14,Kajal,23,MSC
15,Komal,22,Btech

//1. Add Student

```
===== STUDENT MANAGEMENT SYSTEM =====  
1. Add Student  
2. View Students  
3. Search Student  
4. Update Student  
5. Delete Student  
6. Exit  
Enter Your Choice: 1  
Enter ID: 1  
Enter Name: Charu  
Enter Age: 22  
Enter Course: Pharma  
Student Added Successfully!
```

//2.View Student

```
===== STUDENT MANAGEMENT SYSTEM =====  
1. Add Student  
2. View Students  
3. Search Student  
4. Update Student  
5. Delete Student  
6. Exit  
Enter Your Choice: 2  
-----  
ID      : 11  
Name    : Bunny  
Age     : 23  
Course  : MCA  
-----  
ID      : 12  
Name    : Pranali  
Age     : 22  
Course  : MCA  
-----  
ID      : 13  
Name    : Ketaki  
Age     : 21  
Course  : Arch  
-----  
ID      : 14  
Name    : Kajal  
Age     : 23  
Course  : MSC  
-----  
ID      : 15  
Name    : Komal
```

//3. Search Student

```
===== STUDENT MANAGEMENT SYSTEM =====  
1. Add Student  
2. View Students  
3. Search Student  
4. Update Student  
5. Delete Student  
6. Exit  
Enter Your Choice: 3  
Enter Student ID: 14  
-----  
ID      : 14  
Name    : Kajal  
Age     : 23  
Course  : MSC
```

//4. Update Student

```
===== STUDENT MANAGEMENT SYSTEM =====  
1. Add Student  
2. View Students  
3. Search Student  
4. Update Student  
5. Delete Student  
6. Exit  
Enter Your Choice: 4  
Enter Student ID: 15  
Enter New Name: Sarika  
Enter New Age: 22  
Enter New Course: MCA  
Student Updated Successfully!
```

//5. Delete Student

```
===== STUDENT MANAGEMENT SYSTEM =====  
1. Add Student  
2. View Students  
3. Search Student  
4. Update Student  
5. Delete Student  
6. Exit  
Enter Your Choice: 5  
Enter Student ID: 15  
Student Deleted Successfully!
```

//6. Exit

Enter Your Choice: 6

Thank You!

PS C:\Users\prana\OneDrive\Desktop\JAVA PROJECTS\SMSfileBased>